Appendix A3: literature data treatment.

Fishing gears

Exploration of fishery statistics data

Small-scale coastal fishing

We analysed dataset D1, subset to ICES area 27.3.D.30, Swedish country code, vessels categories VL0812 and VL0008 to identify the mostly used métiers by the Small-scale coastal fishing (ssf). A **métier** is defined as group of fishing activities targeting a similar species or assemblage of species, using similar gear (DCF 2021). Figure B.1 shows average total fishing days by métier over the available data (years 2008-2024). We select for consideration in our study the métiers (1) gillnets-freshwater species (GNS-FWS), (2) gillnets-small pelagic species (GNS-SPF), (3) pound nets- anadromous species (FPN-ANA). The sum of average fishing days for the three selected métiers accounts for 85% of total fishing days for ssf in our study area.

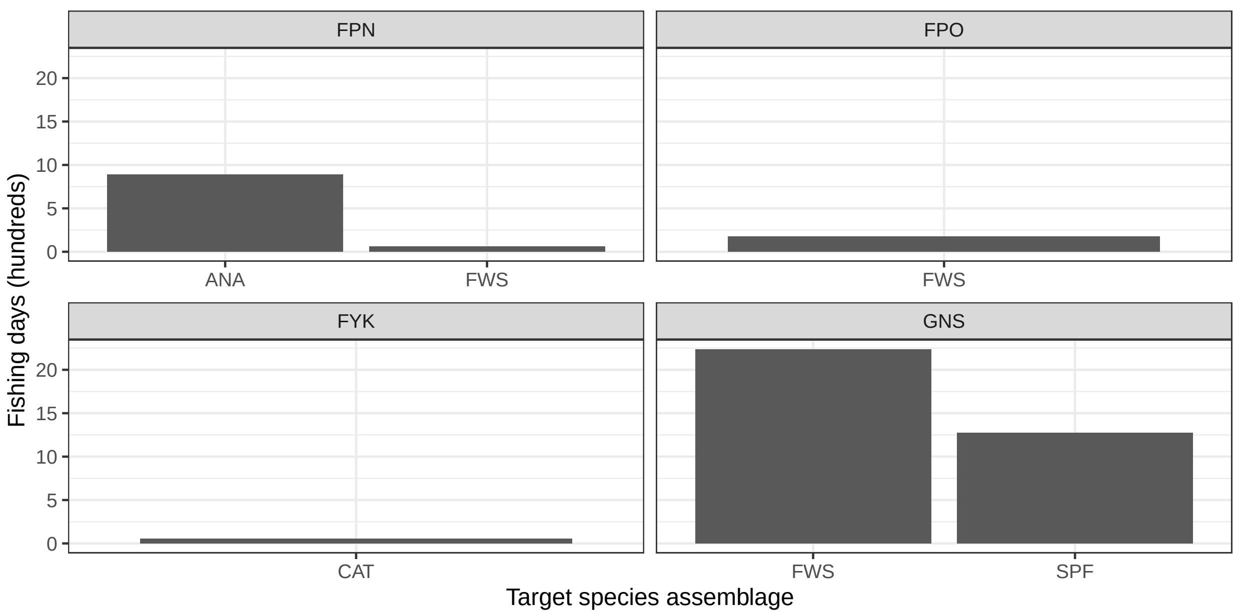


Figure 1: Fishing days by metier, swedish small scale coastal fleet in Bothnian Sea. Panels represents gear type.

Pelagic trawling and large scale fleet

We analysed dataset D1, filtered to Swedish fleet, vessels categories VL2440 and VL40XX, to assess the fishing effort distribution by fishing technique in the Swedish large scale fleet. This information supports the interpretation of economic data, for which data are aggregated across fishing techniques. Figure B.2 shows that the fishing technique *Demersal Trawl Shelf* (DTS) and *Trawl Midwater* (TM) includes similar amount of fishing days, and togheter represents the vast majority of Swedish fishing effort. In the DTS category, pelagic trawling gears (OTM, OTT, PTM) dominates the fishing activity.

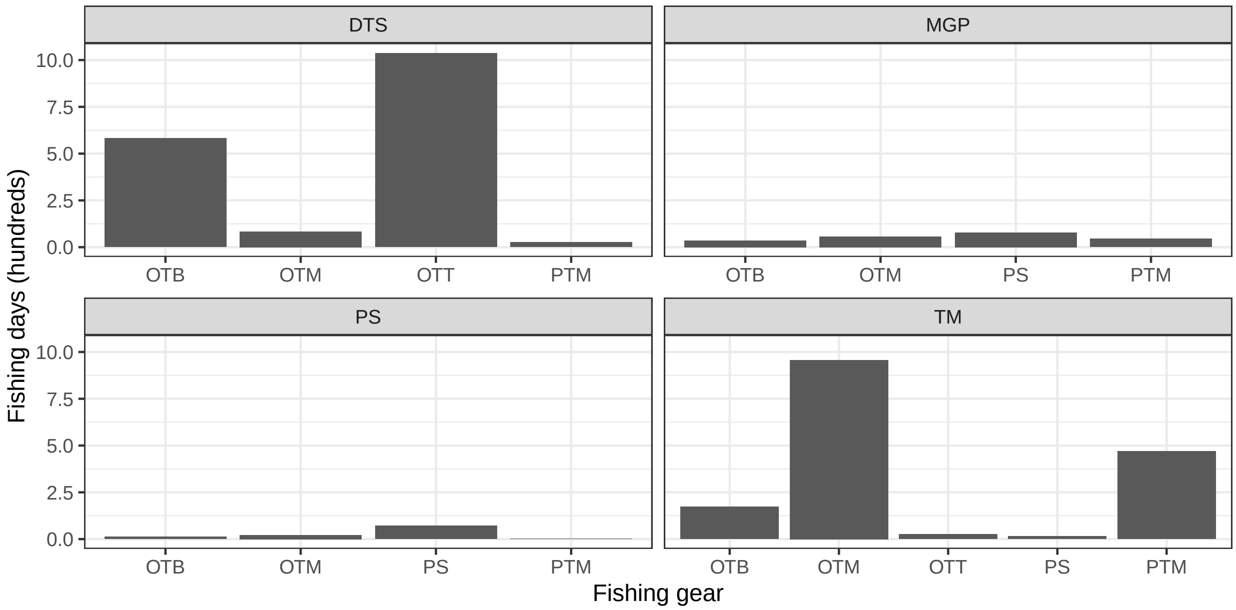
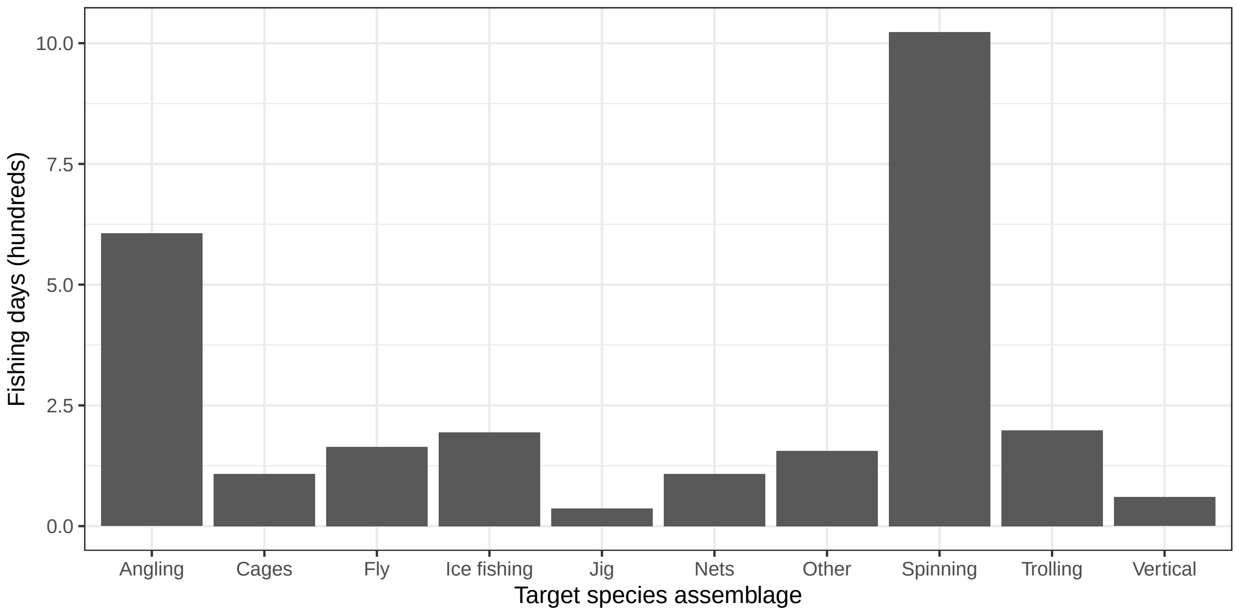


Figure 2: Fishing days by gear and fishing technique, swedish large scale fleet. Panels represents fishing technique.

Recreational fishing

Dataset D4 shows the number of practitioners by recreational fishing method. Fishing methods based on hook and lines are dominating the distribution reported in Fig. B.3. Cages and nets are the two main techniques that do not base on hook and line. Cages are predominantly used in crustaceans fishery on the west coast of Sweden. The use of nets in popular in the Baltic Sea. For the purpose of this study we consider that only hook and line methods apply for the recreational fishing style, with no distinction among methods. Nets applies to the household fishing style.



Assignations

Table 1: Baseline fishing gear assignation

gear	recreational	small_scale	household	trawler	guide
otm	0	0.0	0.0	1	0
gns_fws	0	0.5	0.5	0	0
gns_spf	0	0.3	0.5	0	0
fpn_ana	0	0.2	0.0	0	0
lhp	1	0.0	0.0	0	1
icefishing	0	0.0	0.0	0	0
inland	0	0.0	0.0	0	0

Fishing target

Exploration of fishery statistics

Small-scale coastal fishing

We analysed dataset D2 to identify the catch composition associated to ssf métiers selected in the previous section. The dataset, filtered to Swedish fleet in ICES area 27.3.D.30, vessels categories VL0812 and VL0008, was subset to match the ssf métiers (1) gillnets-freshwater species (GNS-FWS), (2) gillnets-small pelagic species (GNS-SPF), (3) pound nets- anadromous species (FPN-ANA). Species scientific name were assigned according to ASFIS List of Species for Fishery Statistics Purposes (FAO 2026).

Catch data were aggregated by metier and averaged over years. We calculated the cumulative contribution of each species to a metier and we excluded species falling beyond the 95th percentile. Figure C.3 shows average tons by species and metier over the available data (years 2008-2024). The species mostly represented are Atlantic herring (target of gillnets for small pelagic fish), Atlantic salmon (main target of pound nets for anadromous species), European perch (main target of gillnets for freshwater species). Whitefish is also considered a relevant species as it significantly contribute to catches for FPN-ANA and GNS-FWS.

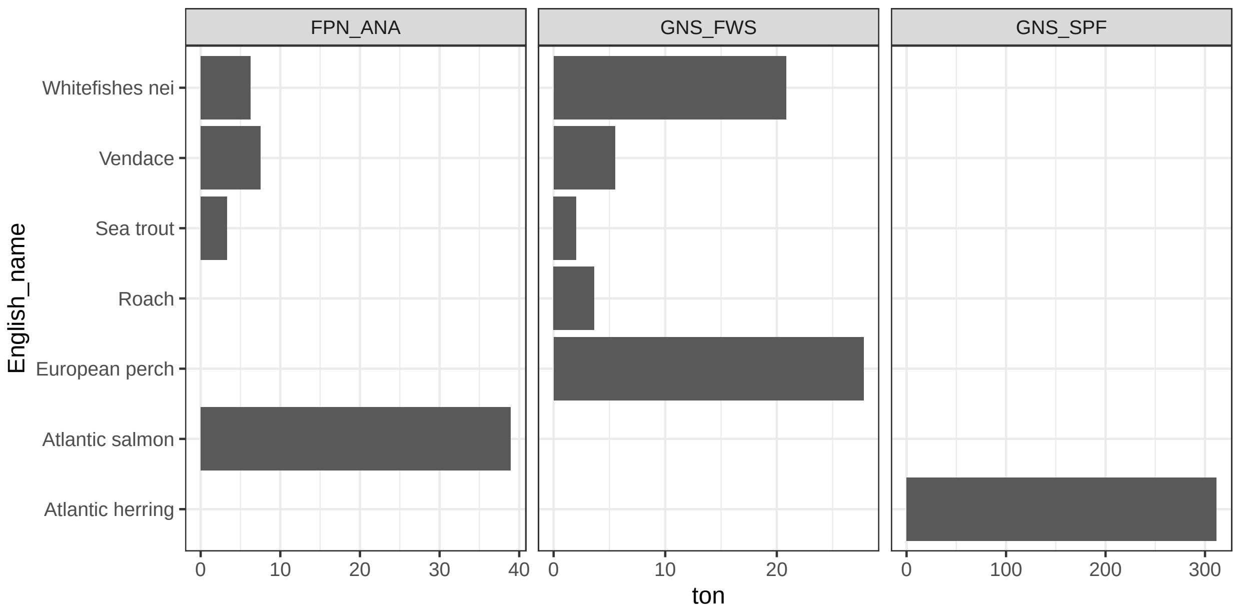


Figure 3: Catches by metier, swedish small scale coastal fleet in Bothnian Sea (selection of metiers)

Pelagic trawling and large scale fleet

We analysed dataset D2 to evaluate the catch value associated with fishing gears deployed by Swedish large scale fleet, and then the catch composition of pelagic trawlers. The economic value of catches, filtered to Swedish fleet in ICES area 27.3.D.30, and vessel categories 'VL2440', 'VL40XX' was aggregated by metier and averaged over years. Figure B.5 shows that pelagic fleet (TM) dominates the distribution (58% of total), and the fishing gears Otter Trawl Midwater (OTM) contributes nearly to 37% of the entire landed value.

To explore fishing target of the pelagic trawl fleet in the study area, dataset D2 was subset to match the gear type "OTM", Swedish fleet in ICES area 27.3.D.30, and vessel categories 'VL2440', 'VL40XX'. Species scientific name were assigned according to ASFIS List of Species for Fishery Statistics Purposes (FAO 2026). We calculated the cumulative contribution of each species to a metier and we excluded species falling beyond the 99th percentile. Figure B.6 shows that Atlantic herring contributes for more than 97% of average catches. For pelagic trawlers we therefore consider Herring as the only relevant species.

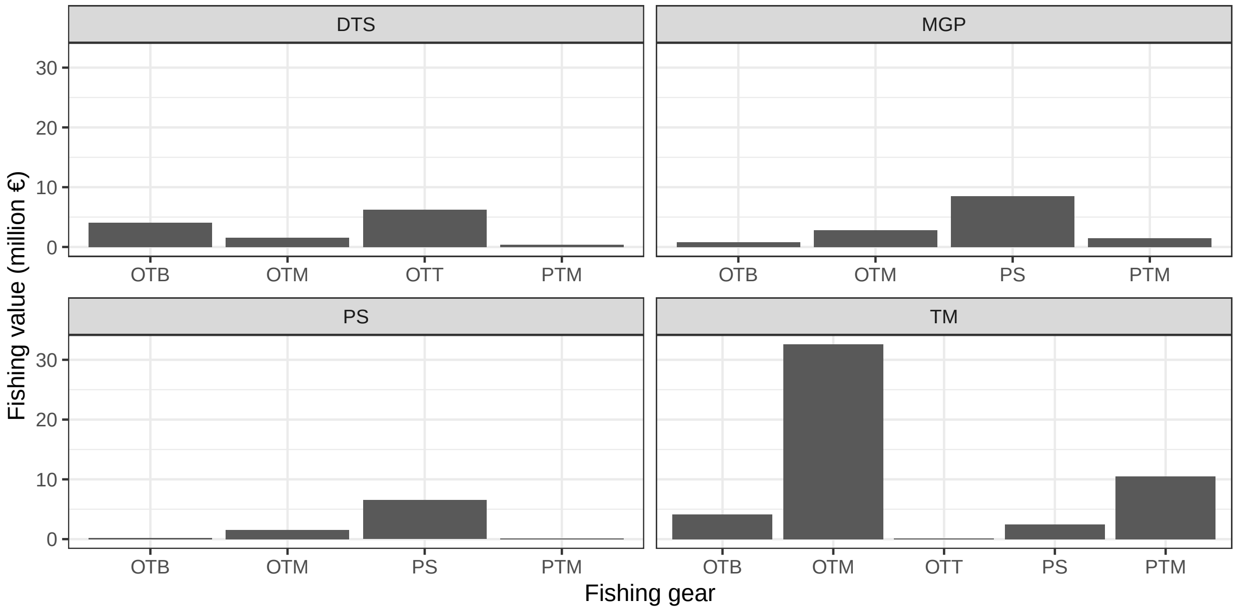


Figure 4: Catches by gear and fishing method, swedish large scale fleet

Recreational fishing

For recreational fishers we are interested in the distribution of number of practitioners by target species, in the Sea. Dataset D5 was subset to exclude species that are not available in the Baltic Sea. Fig. B.7 shows that Perch and pike dominates the distribution. We excluded some relevant species (Cod, Zander) for simplifyng the model. Whitefish was included, although not contributing substantially to recreational fishing target, because it is a shared resource with small scale fishers.

Assignations

Table 2: Baseline fishing gear assignation

target	otm	gns_fws	gns_spf	fpn_ana	lhp	icefishing	inland
herring	1	0.00	0.99	0.00	0.07	0.00	0
perch	0	0.44	0.00	0.00	0.29	0.33	0
pike	0	0.00	0.00	0.00	0.18	0.33	0
salmon	0	0.00	0.00	0.66	0.05	0.00	0
sea_trout	0	0.03	0.00	0.06	0.13	0.00	0
whitefish	0	0.42	0.00	0.23	0.03	0.33	0
not_specified	0	0.10	0.01	0.05	0.25	0.00	1

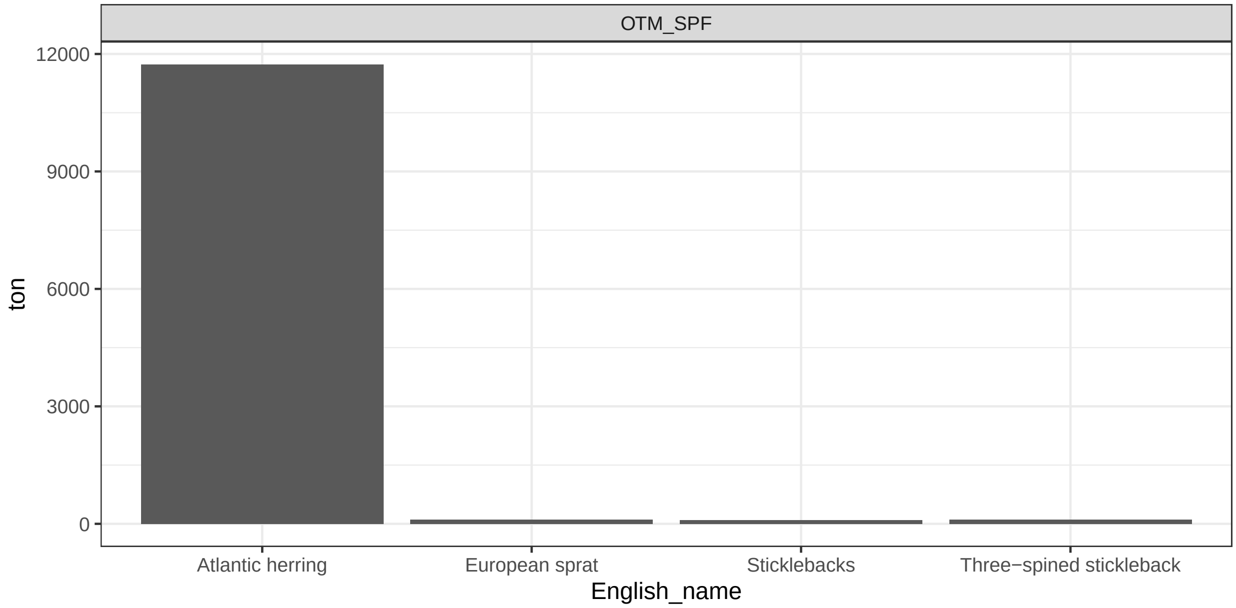


Figure 5: Catches for pelagic trawlers, swedish fleet in Bothnian Sea

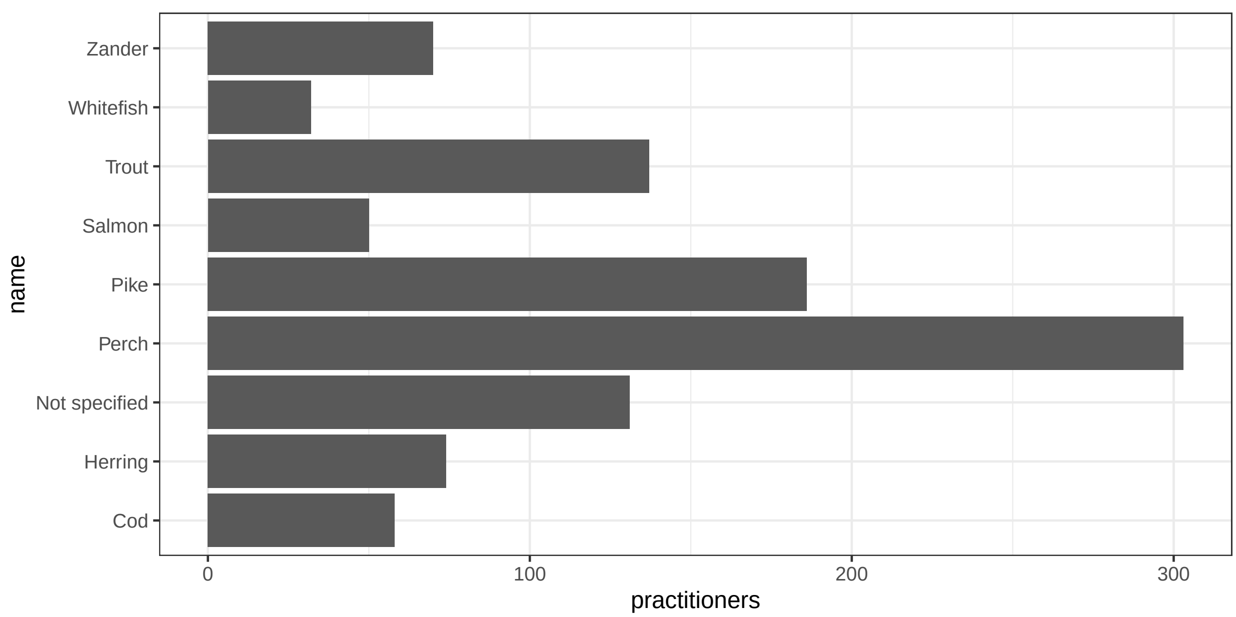


Figure 6: Catches for recreational fishers

DCF. 2021. "Fishing Activity - Metier - Data Collection Framework - DCF." https://dcf.ec.europa.eu/data-calls/definitions-and-terminology/f/fishing-activity-metier_en.

FAO. 2026. "ASFIS List of Species for Fishery Statistics Purposes - All Information Collections." https://www.fao.org/fishery/en/collection/asfis.